

Block 00b — first-review remediation receipt

Status: **implementation and local remediation checks complete; hosted final-head qualification pending at report generation.** The original external review remains “Changes recommended.” No second review was requested. This is not audit acceptance, release approval, permission to merge, or a claim that all possible defects have been found.

Identity and scope

- Owner: Codex /root, keepkey-vault-v11-agent-3; resume existing 715-00b claim.
- Worktree/branch: /private/tmp/kk-715-00b-reconciled, audit/715-00b-reconciled.
- Code-bearing PR: [#856](#), targeting accepted 00a release/715-stack-00-audit-base at 9cb72384fd14b2a688a70d17711a7923941d2290.
- Reviewed source: d2b1cdfb15072112653c7314d898c612dd7df86f, review 5297484301 (20 comments; the review body repeats these and adds no separate concrete finding).
- Remediation code: 6e49ed5dd7794888e9514b0531a32a5c9eabf782. Reports added afterward change documentation only. The containing final report head and hosted CI run are attached to the PR body.
- Main ledger SHA256 at resumed intake: d4ab607b00d5b0db1b3ae2cfc689e921265f4448419cc9a4e165f2d234f99275; SOP: 84885757310cdd6f633261e09d71e6207a0b6406dba507e0db0603cb84f5bdf2.
- Pins unchanged: Python 98c717ff2204124bf67cc78cab8ca1d501fd4e92; protocol dd9c85dc747cf965fb0e7bf49615dc9e7568ee65; crypto 74908938c562ec89dcae9fb4e3b05ec3ff17f58b.

The existing PR is a broad integration surface, a recorded exception to the SOP’s small-unit preference. This remediation reviews the reported contracts and affected downstream paths in four groups: lifecycle/concurrency; signing consent/validation; recovery/storage; build/test execution. The prior receipt is historical evidence, superseded by this receipt for current status. This pass does not independently requalify every unchanged feature in the original 108-path replay. Further feature extraction and assembled-product acceptance remain separate batch work; no green-test total is offered as a substitute.

Individual review dispositions

Each ID below is the GitHub inline-comment ID. Source evidence is the frozen code commit above. “Fixed” means the implementation and listed local assertion pass; it does not mean Copilot reviewed the fix. All findings originate in the reviewed integration candidate. The shutdown pair is one root cause. The disposition replies are posted only after the fix commit is pushed.

Comment ID	Disposition	Contract and source change	Executed evidence
4087896292	Fixed, same root as 6361	kkemu_shutdown calls stop/join before abort, storage commit, ring clearing or flash release.	EmulatorLifecycle.ShutdownStopsPollThreadAndAllowsRestart, ShutdownWakesConfirmationWaitingForHostDecision.
4087896331	Fixed	Define thorchain_isValidAsset; apply the declared grammar before deposit display and in the serializer.	Thorchain.DeclaredDepositAssetValidatorEnforcesGrammar, DepositAssetAndSignerFailClosed; full native and ARM.
4087896361	Fixed, duplicate of 6292	Same lifecycle correction; retained as a separate thread disposition.	Same two real-library lifecycle tests.
4087896390	Fixed	Producer drops the new frame when full, never writes the consumer index or unread slot; dropped frames remain eligible for retry. Acquire/release publishes completed slots.	All four EmulatorLifecycle tests; old producer fails OverflowPreservesUnreadFramesAndRetriesDroppedFrame.
4087896411	Fixed	Complete UA text uses measured byte paging, followed by the address-specific helper that invokes the actual QR layout.	ZcashDisplay.FullAddressPagesThenQrAndCancellation: multiple text pages, real QR callback with the entire address, rejection at final text page and QR.

Comment ID	Disposition	Contract and source change	Executed evidence
4087896441	Refuted for signing entry path	ethereum_contractHandled first requires total length equal to live chunk length; zx_isZxApproveLiquid then requires exactly 68 bytes. The alleged streamed-tail route cannot reach specialized consent.	Ethereum.ApproveLiquidityRouterRejectsUnreviewedTail: exact 68 accepted by router; 69/1024 totals with 68-byte chunk rejected; complete 69-byte chunk rejected.
4087896493	Fixed	MAYA asset grammar is enforced by the handler before approval and by the public deposit serializer.	ReviewHandlers.MayaDepositGrammarRejectedBeforeConsent, Mayachain.DepositAssetAndSignerFailClosed, AssetGrammarRejectsSafeTextOutsideContract.
4087896520	Fixed	Missing/empty TRON signing messages fail before consent or derivation; direct-call lengths are bounded. Verify also refuses empty input.	ReviewHandlers.MissingAndEmptyTronMessageNeverRequestsConsent; original handler fails negative control.
4087896558	Fixed	Every signed TRON byte is reviewed using confirm_bytes; no 114-character/57-byte prefix copy remains.	ReviewHandlers.RejectTronSignedAndVerifiedMessageTail, printable and binary 200-byte messages.
4087896606	Fixed	Verification displays the full authenticated message after signer approval; tail rejection cannot report success.	Same test builds a valid signature, verifies it, then rejects the last message page. Original prefix implementation fails.
4087896650	Partially refuted; grammar hardened	Original public serializer already rejects NULL, quotes and controls through tendermint_validateSafeText. It now also enforces the documented denomination alphabet, rejecting safe-text punctuation such as colon/underscore.	Mayachain.SendSerializerRefusesInvalidDenomWithoutConsumingMessage; malformed inputs leave the one-message budget available for a valid send.
4087896671	Fixed conservatively	More than eight colon-separated fields returns UNPARSED before any specialized approval; callers then disclose the complete raw memo.	Mayachain.MemoFieldCapacityFallsBackBeforeAnyApproval, SWAP and ADD. Old parser fails.
4087896706	Fixed	Reset wipes previous mnemonic word and its formatted display equivalent; display scratch is also wiped immediately after rendering.	Recovery.AbortAndResetClearPreviousWordAndDisplayEquivalent in full/BTC. Removing wipes makes it fail.
4087896746	Fixed	Include guarded Windows and BCrypt declarations in the RNG translation unit.	Installed x86_64-w64-mingw32-gcc -fsyntax-only compiles rng.c with EMULATOR and RAND_PLATFORM_INDEPENDENT. Full Windows runtime testing is not claimed.
4087896778	Fixed	Clear/check errno around decimal strtoll, before consent or hashing; remove obsolete tiny-json header declaration that conflicts with newlib errno.	Eip712.DecimalOverflowRefusedWithoutHashMutation, DecimalInt64BoundaryMatchesIndependentEncoding; old parser fails. ARM full builds under Werror.
4087896807	Fixed	Common mutation guard refuses newer-format wallet operations; recovery start is covered too. Only explicit confirmed wipe clears the lock.	ReviewHandlers.NewerWalletRefusesEveryMutationBeforeConsent, both variants: seven handlers, unchanged flash/no ceremony, declined wipe preserves flash, accepted wipe clears lock.
4087896829	Fixed	Omitted and empty MAYA send denomination consistently default to cacao for validation, amount display and serialization.	ReviewHandlers.MayaDefaultDenomReachesConsentForMissingAndEmptyField; amount and serializer tests cover cacao semantics.
4087896862	Refuted; adjacent real defect fixed	XRPLF defines MemoData as Blob type 7, field 13: 0x7D is correct. Boundary testing found the distinct 192-byte length-prefix bug; single-byte encoding now includes 192.	Ripple.MemoDataUsesCanonicalBlob13AndLengthBoundary checks canonical markers and bytes at 191/192/193/199; old length boundary fails.
4087896881	Technical decline under existing owner decision	Preserve the owner-required pre-foundation bootloader source restoration. Explicitly distinguish its four-line adjacent diff from zero diff against policy baseline; default release builds still exclude bootloader-family artifacts.	Direct file equals fc1e93746132553ad98ed60f4847c8d770732bf9; full/BTC clean ARM outputs contain no bootloader/updater/bootstrap artifacts.
4087896910	Fixed, with expanded registration audit	Full emulator xunit/CTest now execute Zcash crypto/display tests, plus previously omitted active TRON and signed-metadata suites. xunit depends on executable targets. Existing XML glob collects new reports.	Full: 349 firmware, 20 board, 19 crypto, 59 Zcash, 1 Zcash display; BTC: 97/20/19. Dedicated macOS CI step runs 4 emulator lifecycle tests.

The XRPL oracle is [XRPLF ripple-binary-codec definitions](#): MemoData has nth: 13, type Blob. This is independent of the firmware serializer. Do not change 0x7D to the reviewer's proposed 0x72.

Additional findings exposed by executing the missing suites

1. **Native value consent:** v2 metadata matching computed metadata_schema_moves_value, but Ethereum discarded it and suppressed the amount screen. The caller now preserves that confirmation; matching/reset clear stale state. SignedMetadataTest.PayableSchemaCannotSkipNativeAmountConsent fails when the caller is restored to its old behavior. V2SchemaPreservesNativeValueConfirmation checks nonzero, failed rematch and zero transitions.
2. **Dormant crypto build drift:** Zcash tests used the old four-argument RedPallas signing API and omitted the pinned pallas_ct.c implementation. Tests now provide an explicit 80-byte randomness fixture and link the required implementation. No production signing randomness is replaced.
3. **Dormant fixture drift:** runtime-signer tests omitted the existing AdvancedMode precondition; their fixture now establishes it, while a separate negative test proves off-mode rejection. Two old expectations conflicted with current supported behavior: payable schemas retain native-value review, and fixed-word BYTES is supported. FixedBytesSchemaDecodesEntireWord checks all 32 decoded bytes and rejects the second display page. Unsupported dynamic STRING remains rejected.
4. **Build portability:** move shared-library links to the target's defining CMake directory so the declared minimum CMake does not need the newer cross-directory-link policy. GCC diagnostic builds exercise C11 atomics; the older container Clang lacks stdatomic.h for the shared library. Ordinary Clang full/BTC and GCC shared-library builds are recorded separately.

Security closure matrix and adversarial pass

For the following rows, the hostile origin is the host/protocol unless stated otherwise. The observer is the device user for consent, the host for result messages, and a later in-process observer for residual memory. Native tests execute public handlers or the named engine entry; white-box probes are confined to test binaries and are never firmware secret getters. Full/BTC differences below are explicit exclusions, not implicit passes.

Value/state; origin; phase; variant	Allowed output	Forbidden output/ equivalent	Entry, consumer, counterexample
Emulator producer state; library lifecycle; init/poll/stop/ shutdown/restart; host	Joined thread before teardown; complete ordered captured frames	Live producer against cleared rings/ flash; torn/ overwritten unread slot	Public init/start/shutdown and real display callback; ring saturation, retry, concurrent producer/consumer, parked Ping confirmation. Signal-driven timer scheduling is not a race-detector proof; test drives lifecycle with explicit clock control.
TRON message; sign/verify; full	Every signed byte, including binary escaped bytes and suffix, approved	Signature/success after last-page refusal; empty-message signature	Real FSM handlers; 200 printable bytes and 200 binary bytes, independent valid signature before verify.
MAYA/THOR asset and denom; validation/approval/ hash; full	Valid chain alphabet and the same resolved cacao value at display/ hash	NULL dereference, punctuation outside grammar, contradictory default	Handler and public serializer tests; colon/underscore/control encodings, missing/empty field, valid send after invalid inputs.
MAYA memo; parse/ approve; full	Structured complete memo or full raw review	CONFIRMED after silently dropping extra structure	SWAP/ADD with ninth field; negative control restores old bounded split. Standard measured renderer fails closed when a single field cannot fit.
EIP-712 decimal; parser/consent/hash; full	Exact supported signed 64-bit value, sign-extended to 256 bits	Saturated LLONG value replacing requested input	Int64 extrema with independently built bytes; overflow/underflow and huge decimal leave hash context unchanged and consume no consent. Wider integer support and public typed-signing policy are not expanded.
Runtime metadata/ value; matching/ consent; full	Authenticated complete bytes, native transfer screen, user-enabled runtime trust	Hidden native currency; second half of fixed bytes omitted; stale value flag	Direct ethereum_signing_init with payable v2 schema; decline after five metadata and native-value screens; BYTES second-page rejection; AdvancedMode off negative.

Value/state; origin; phase; variant	Allowed output	Forbidden output/ equivalent	Entry, consumer, counterexample
Recovery mnemonic and formatted previous-word string; abort/reset; both	Cleared scratch; later ceremony starts without prior word	Word recoverable from static previous-word or formatted equivalent	Real reset implementation compiled into test probe; fill both representations, setup_abort, reset and repeated abort. This is memory cleanup evidence, not a new confidentiality claim about all recovery screens.
Unknown-format flash wallet; boot/mutate/ wipe; both	Non-destructive refusal; erase only after wipe approval	Success for an unpersisted mutation; implicit wallet erase	Seed actual emulated storage sector with newer version; seven handlers, flash equality, decline and accept wipe.
Zcash UA; display/ approval/cancel; diagnostic full	Complete text pages followed by actual QR callback with same full string	Prefix-only text or helper ignoring QR callback	Compile real app_confirm/app_layout guarded code in separate test binary; text-tail and QR rejection. Handlers remain disabled in this foundation; no device scan or shielded signing acceptance claimed.
Windows random bytes; platform RNG; emulator	Declared BCrypt system RNG entry and existing checked failure	Undeclared identifiers in translation unit	MinGW syntax compile; no Windows runtime execution claim.
Calldata extent and XRPL memo fields; full	Exact approved fixed extent; canonical independent field/ length encoding	Stream tail silently specialized; wrong field/192-byte prefix	Router predicates exercised at 68/69/1024; independent XRPL markers at length boundaries.
Test assertions; build/ CI; full/BTC/ diagnostic	Executed named suites and emitted XML	Test file/target existence presented as execution	xunit dependencies/commands and CTest registration; actual XML inventories. Hive is absent from this foundation's implementation source list and is not claimed tested. Zcash engine/display are diagnostic; firmware handlers remain guarded off.

The separate falsification pass checked the consumers of the fixes, not just local edits. It caught the unused native-value flag and the incorrect 192-byte prefix. Isolated container overlays restored the defects without changing the candidate worktree: **9 of 12 selected firmware tests failed**, while positive/default-path controls continued to pass; the original frame producer failed its overflow regression. Restoring the candidate produced passing suites. Native denial tests assert failure status and unused consent pairs, so an early unrelated refusal cannot masquerade as successful tail coverage.

C-2 policy reconciliation

No bootloader behavior change is authorized. The file `tools/bootloader/usb_flash.c` is unchanged by this remediation and equals the owner-selected pre-foundation policy baseline `fc1e93746132553ad98ed60f4847c8d770732bf9`. Against accepted 00a it contains a four-line restoration of the `DEBUG_LINK` watermark block, because 00a removed that block. Therefore “zero adjacent bootloader source changes” would be false. Removing the restoration would contradict the existing explicit owner instruction documented in the original receipt. This technical decline preserves that instruction and makes both comparisons reviewable. Full/BTC default ARM builds emit no bootloader, updater or bootstrap artifacts. Opt-in bootloader builds and release publication remain outside scope.

Verification and limits

Check	Result	Configuration / artifact
Full ordinary xunit	349 firmware, 20 board, 19 crypto, 59 Zcash crypto, 1 Zcash display passed	Clang container; all five XML files included by report glob.
Bitcoin-only xunit	97 firmware, 20 board, 19 crypto passed	Same toolchain, <code>KK_BITCOIN_ONLY=ON</code> ; non-BTC chain tests excluded.
Threaded library	4 lifecycle/concurrency tests passed	<code>GCC</code> , <code>KK_BUILD_DYLIB=ON</code> ; macOS CI executes same target.
Full OLED / host	64 passed, 23 skipped OLED; 559 passed, 272 skipped host	Declared 7.14.3 foundation capability profile, exact Python pin; screenshots audited.
BTC OLED / host	25 passed, 62 skipped OLED; 334 passed, 497 skipped host	Exact BTC emulator; screenshots audited.
Report validator	7 passed, both host environments	<code>test_report_variant_validation.py</code> .

Check	Result	Configuration / artifact
ARM full / BTC	Both MinSizeRel builds and SRAM gates passed	Reserves 21,736 / 31,368 B; largest frame 7,664 B; no forbidden bootloader artifacts.
Windows RNG	Syntax compile passed	Installed MinGW GCC; not a complete Windows runtime qualification.
Mechanical preflight	Diff whitespace, actionlint, code-bearing gate pass	Actual accepted 00a predecessor; pins unchanged.
Hosted final head	Pending at generation; attach exact-head run to PR	Earlier d2b1cdf CI cannot qualify changed firmware.

Evidence inventories and SHA256s are in 715-00b-remediation-evidence-20260923.json. Logs live under /private/tmp/kk00b-remediation-* and /private/tmp/kk00b-host-final-*. Hosted CI publishes durable artifacts after push. Capability skips remain individually represented in JUnit; none are counted as passing assertions. Physical signing/OLED, physical power interruption and signed-device upgrade remain release gates. Upstream crypto PR #12 remains a publication dependency.

Implementation, targeted checks and local affected integration are complete for these dispositions. Adversarial evidence has the limits stated above. External review delivery is historical; there is no fresh clean review. Finding dispositions are ready for replies after push. Audit acceptance and release acceptance remain pending. Continue the batch from the canonical main ledger; do not advance a dependent block as though 00b were accepted.

Frozen code line inventory

The adjacent inventory below is 9cb72384fd14b2a688a70d17711a7923941d2290..6e49ed5dd7794888e9514b0531a32a5c9eabf782. It includes the historical source/PDF receipt and every implementation, test, pin and workflow path in that code-bearing range. This new remediation source/PDF and evidence JSON are added in the following documentation receipt commit; their final counts and containing SHA are recorded in the PR body. The code delta itself is 39 paths, 877 additions and 168 deletions, grouped by the contract tables above. Do not sum replay history as a net candidate diff.

Path	Added	Deleted
.github/workflows/ci.yml	22	0
.gitmodules	2	2
CMakeLists.txt	56	11
cmake/toolchains/mingw-w64-x86_64.cmake	40	0
deps/crypto/CMakeLists.txt	13	0
deps/crypto/trezor-firmware	1	1
deps/device-protocol	1	1
docs/release/audit-units/715-00b-review-20260923.md	243	0
docs/release/audit-units/715-00b-review-20260923.pdf	-	-
docs/release/audit-units/715-00b-storage-persistence.md	92	0
include/keepkey/board/bsd_compat.h	28	0
include/keepkey/board/draw.h	25	0
include/keepkey/board/layout.h	18	0
include/keepkey/board/messages.h	6	0
include/keepkey/emulator/libkkemu.h	61	17
include/keepkey/firmware/app_confirm.h	3	0
include/keepkey/firmware/app_layout.h	9	1
include/keepkey/firmware/bip85.h	22	0
include/keepkey/firmware/eip712.h	2	9
include/keepkey/firmware/ethereum.h	1	0
include/keepkey/firmware/ethereum_contracts/thortx.h	27	1
include/keepkey/firmware/ethereum_tokens.h	4	0

Path	Added	Deleted
include/keepkey/firmware/fsm.h	22	0
include/keepkey/firmware/hive.h	255	0
include/keepkey/firmware/mayachain.h	9	0
include/keepkey/firmware/osmosis.h	16	0
include/keepkey/firmware/signed_metadata.h	240	0
include/keepkey/firmware/solana.h	151	8
include/keepkey/firmware/storage.h	7	0
include/keepkey/firmware/tendermint.h	4	0
include/keepkey/firmware/thorchain.h	19	1
include/keepkey/firmware/tiny-json.h	0	1
include/keepkey/firmware/tron.h	55	0
include/keepkey/firmware/zcash.h	384	0
include/keepkey/transport/interface.h	2	0
include/keepkey/transport/messages-ethereum.options	25	0
include/keepkey/transport/messages-hive.options	58	0
include/keepkey/transport/messages-solana.options	4	0
include/keepkey/transport/messages-zcash.options	54	0
lib/board/draw.c	82	3
lib/board/layout.c	162	3
lib/board/messages.c	90	40
lib/board/timer.c	28	2
lib/board/udp.c	6	0
lib/board/usb.c	29	20
lib/emulator/CMakeLists.txt	34	0
lib/emulator/libkkemu.c	271	34
lib/emulator/setup.c	31	2
lib/emulator/udp.c	8	2
lib/firmware/CMakeLists.txt	1	0
lib/firmware/app_confirm.c	24	0
lib/firmware/app_layout.c	89	3
lib/firmware/bip85.c	105	0
lib/firmware/eip712.c	11	2
lib/firmware/eos.c	6	0
lib/firmware/ethereum.c	128	33
lib/firmware/ethereum_contracts.c	2	0
lib/firmware/ethereum_contracts/thortx.c	169	172
lib/firmware/ethereum_contracts/zxapliquid.c	2	1
lib/firmware/ethereum_contracts/zxliquidtx.c	11	6
lib/firmware/fsm.c	38	0
lib/firmware/fsm_msg_bip85.h	142	0
lib/firmware/fsm_msg_common.h	12	0
lib/firmware/fsm_msg_ethereum.h	138	0
lib/firmware/fsm_msg_hive.h	991	0
lib/firmware/fsm_msg_mayachain.h	34	6

Path	Added	Deleted
lib/firmware/fsm_msg_ripple.h	15	0
lib/firmware/fsm_msg_solana.h	655	165
lib/firmware/fsm_msg_thorchain.h	24	3
lib/firmware/fsm_msg_tron.h	125	75
lib/firmware/fsm_msg_zcash.h	1414	0
lib/firmware/hive.c	1016	0
lib/firmware/mayachain.c	95	80
lib/firmware/messagemap.def	6	0
lib/firmware/osmosis.c	18	0
lib/firmware/recovery_cipher.c	62	5
lib/firmware/reset.c	21	7
lib/firmware/ripple.c	20	1
lib/firmware/signed_metadata.c	1067	0
lib/firmware/solana.c	568	291
lib/firmware/storage.c	65	7
lib/firmware/storage.h	5	0
lib/firmware/tendermint.c	28	0
lib/firmware/thorchain.c	43	9
lib/firmware/tiny-json.c	5	3
lib/firmware/tron.c	359	0
lib/firmware/zcash.c	1133	0
lib/rand/rng.c	14	0
lib/transport/CMakeLists.txt	18	0
scripts/build/docker/device/release.sh	6	1
scripts/emulator/docker-compose.bitcoin-only.yml	2	0
scripts/emulator/docker-compose.yml	9	0
scripts/emulator/firmware-unit.sh	8	0
scripts/generate-test-report.py	71	6
tools/CMakeLists.txt	5	3
tools/bootloader/usb_flash.c	4	0
tools/emulator/CMakeLists.txt	11	13
tools/firmware/CMakeLists.txt	3	1
unittests/crypto/CMakeLists.txt	17	0
unittests/emulator/lifecycle.cpp	110	0
unittests/firmware/CMakeLists.txt	15	0
unittests/firmware/coins.cpp	11	0
unittests/firmware/ethereum.cpp	18	32
unittests/firmware/hive.cpp	783	0
unittests/firmware/mayachain.cpp	48	0
unittests/firmware/recovery.cpp	18	0
unittests/firmware/recovery_review_probe.c	18	0
unittests/firmware/review_handlers.cpp	216	0
unittests/firmware/review_integer.cpp	58	0
unittests/firmware/ripple.cpp	27	0

Path	Added	Deleted
unittests/firmware/signed_metadata.cpp	1633	0
unittests/firmware/solana.cpp	905	505
unittests/firmware/storage.cpp	107	4
unittests/firmware/thorchain.cpp	82	18
unittests/firmware/tron.cpp	495	0
unittests/firmware/zcash.cpp	1768	0
unittests/firmware/zcash_display.cpp	43	0
unittests/firmware/zcash_display_confirm_probe.c	24	0
unittests/firmware/zcash_display_layout_probe.c	3	0